01 - Follow a message through a network

Explained answers and feedback

Separate local forwarding, routing and application responsibilities.

Textbook: networks

A

Computer A -> switch -> computer B. Same-subnet delivery uses local link forwarding after address resolution; a router is needed when the destination is outside the local subnet.

B

A cached page may require no new DNS lookup or network fetch. Request a supplied DNS result, connectivity observation, or server response. Consider alternatives instead of asserting a fault from the cache alone.

Feedback

If every box is called a router, label each forwarding responsibility and revisit the simplified path. Real devices can combine roles.

02 - Compare transport using a supplied trace

Explained answers and feedback

Explain reliability and latency trade-offs without treating protocols as guarantees.

Textbook: networks

A

Next expected byte 206 after the first; 212 after both contiguous segments. Missing bytes 206..211 still need recovery. This simplified exercise counts payload bytes only.

B

Report: TCP suits ordered reliable transfer, with retransmission delay as a cost. Voice: UDP may suit time-sensitive delivery; loss/order handling falls to the application. Applications can add acknowledgements/retries themselves.

Feedback / exit

No. Congestion, temporary interference or incomplete capture may explain a missing observation. Request repeat evidence or relevant link/error logs before selecting a physical fault.

03 - Design against capacity and access requirements

Explained answers and feedback

Calculate a lower-bound transfer time and justify a restricted network design.

Textbook: network-design

A

75 MB * 8 = 600 Mbit; 600 / 20 = 30 seconds ideal minimum. Accept contention, overhead, wireless interference, storage or other identified bottlenecks with explanation.

B

Accept a coherent topology with separate segments and an explicit firewall/access policy at the routing boundary. Permit student -> named learning service; deny student -> staff file service. Test the allowed destination succeeds and the denied one is blocked, retaining observations.

Feedback / exit

An allowed test does not exercise the denied path. Require a corresponding denied-case test; network names or colour coding are not enforcement evidence.

04 - Year 12 bridge: allocate a small subnet

Explained answers and feedback

Calculate an IPv4 subnet range and include every required interface.

Textbook: unit3-addressing

A

Block size 32; network 192.0.2.96; broadcast .127; usable .97-.126; mask 255.255.255.224; 30 usable.

B

31 interfaces do not fit 30 usable. /26 has 64 total and 62 usable. 192.0.2.128/26 spans .128-.191; usable .129-.190; broadcast .191.

Feedback / exit

.95 is the broadcast of .64/27, not a host in .96/27. If allocations overlap, list network and broadcast before assigning host addresses.